

Security Risk Assessment

Grounded in: BD Alaris™ Infusion System
Basis: AAMI SW96 / ISO 14971 structure; illustrative register.

Device	BD Alaris™ Infusion System
Method	AAMI SW96 / ISO 14971-aligned security risk assessment (STRIDE threats)
Note	Illustrative rows showing the expected structure and reasoning, not BD's confidential file.

Security risk register (excerpt)

ID	Asset	Threat (STRIDE)	Exploit.	Patient harm	Control	Residual
S-01	EMR interface	Spoofing	Med	Med — wrong auto-program	Mutual auth, validated orders	Low (accepted)
S-02	Software update	Tampering	Low	High — malicious image	Code signing, integrity check	Low
S-03	Service iface	Elevation	Med	Med	RBAC, least privilege	Low
S-04	PHI in transit	Info disclosure	Med	Low	TLS, key management	Low
S-05	Network	Denial of service	Med	Med — delayed therapy	Safe-state, local fallback	Low (accepted)

Risk acceptance

Residual risks S-01 and S-05 accepted with compensating controls (network segmentation, facility monitoring) and reflected in labeling. Acceptance signed by Security and Quality with rationale recorded in the risk file.

If a control cannot be implemented

Document a risk-based justification: why the control is infeasible, the compensating controls, the residual risk and its formal acceptance, and the user-facing labeling that describes the limitation — consistent with FDA's premarket cybersecurity guidance.

Sources (public)

- FDA / BD — “BD Receives FDA 510(k) Clearance for Updated BD Alaris Infusion System” (Jul 21, 2023).
- FDA — Cybersecurity in Medical Devices: QMS Considerations and Content of Premarket Submissions (final, Jun 27, 2025).
- AAMI SW96; ISO 14971; IEC 81001-5-1.